University of Texas
MSBA
RM 294 - Optimization I

Project 2 Report

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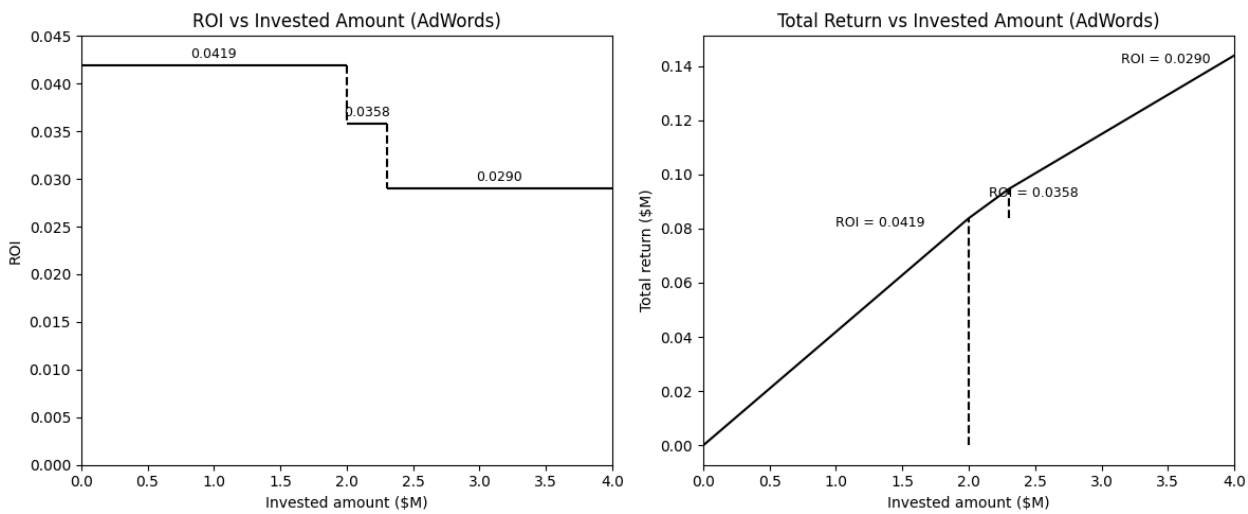
Part 1 - Problem Setup and Data Overview

In this project, our company is allocating a $10 million marketing budget across multiple advertising platforms with the goal to maximize total return on investment (ROI). The marketing mediums include both traditional channels (TV, Print, Email) and digital platforms (Facebook, Instagram, Linkedin, Snapchat, Twitter, SEO< and AdWords). The first consulting firm provided ROI estimates (`roi_company1.csv`) representing piecewise-constant ROI functions for each platform and spending tier. Each platform is divided into tiers corresponding to an investment range and a specific ROI value. These returns are non-increasing, meaning that as higher spending levels are reached, they will yield lower incremental returns. Figure 1 displays the ROI piecewise linear function of AdWords advertising.

Figure 1:



Part 2 - Constraints

In addition to our $10 million budget, our boss has imposed multiple constraints reflecting realistic business priorities and balance requirements. These constraints ensure that our recommendation remains consistent with the company's advertising strategy and guidelines. The following are the constraints provided:

- $10 million marketing budget

$$\sum_{p,t} x_{p,t} \leq 10$$

- The amount invested in each platform must be less than or equal to $3 million

$$\text{For each platform } p: \quad \sum_{t} x_{p,t} \leq 3$$

- The amount invested in print and TV must be less than or equal to the amount invested in Facebook and Email

$$\text{Print} + \text{TV} \leq \text{Facebook} + \text{Email}$$

- The amount invested in social media must be at least twice the amount invested in SEO and AdWords

$$\text{Social} \geq 2 \times (\text{SEO} + \text{AdWords})$$

These constraints demonstrate the priorities of our marketing company, such as an emphasis on digital channels, a desire to increase interaction through social networks, and maintaining a diverse advertising strategy. Together, these requirements define a feasible region for our solution that balances budget limits and marketing priorities.

Part 3 - Linear Programming Model Formulation

With a set of constraints established, the marketing budget was optimized as a Linear Programming model using Gurobi. Because the first consulting firm's data is non-increasing across tiers, the total return function is concave. This allows the problem to be formulated as a Linear Programming model, where the investment across tiers can be represented by continuous decision variables without the need for integer constraints.

In this model, the decision variable $x_{p,t}$ represents the amount of money (in millions) invested in platform *p* at tier *t*. The lower and upper bounds of each decision variable were determined by the consulting firm's ROI data. The objective of the model is to maximize total expected return, which we calculate as the sum of the ROI for each platform-tier pair multiplied by the amount invested in that tier. This model is subject to the constraints that are outlined prior in part 2. The objective equation can be visualized as:

$$\max \sum_{p,t} (\text{ROI}_{p,t}) \times x_{p,t}$$

Now, the LP model produced an optimal model that maximized total expected return according to the ROI data. The results (Figure 2) provided insight into which platforms and spending levels offered the most beneficial returns. The model allocated the full $10 million marketing budget across four platforms (AdWords, Email, Instagram and TV), achieving an overall ROI of 5.44% and yielding an expected return of $540,000. The solution allocated the maximum platform investment of $3 million into Email, Instagram and TV due to their strong early-tier ROIs, in addition to providing a smaller allocation to AdWords. These investment decisions focus on high-performing channels, while maintaining the strategic diversity required by the marketing company. These results establish a benchmark for evaluating alternative ROI structures and integer-based optimization in the following parts.

Figure 2:

```
================== Optimal Solution ==================
Total Budget Used: $10.00M
Total Expected Return: $0.54M
Overall ROI: 5.44%

Platform Breakdown:
-------------------------------------------------------
AdWords      | Invested: $1.00M | Return: $0.04M | ROI:  4.19%
Email        | Invested: $3.00M | Return: $0.15M | ROI:  4.93%
Instagram    | Invested: $3.00M | Return: $0.17M | ROI:  5.71%
TV           | Invested: $3.00M | Return: $0.18M | ROI:  6.08%
=======================================================
```

Part 4 – Mixed-Integer Programming (MIP) Model for Non-Concave ROI

In this stage, the second consulting firm's ROI data (`roi_company2.csv`) violated the non-increasing assumption used in Parts 1–3, meaning the total return function was no longer concave. Because some tiers offered higher returns at higher spending levels, a linear program could produce unrealistic results that "skip" early tiers. To address this, the problem was reformulated as a Mixed-Integer Program (MIP) by introducing binary activation variables for each platform–tier pair. These variables ensure that a tier can only be activated once all previous tiers are fully funded, creating a more realistic spending structure. The model continues to maximize total expected return subject to the same managerial constraints: a total $10 M budget, a $3 M per-platform cap, `Print + TV ≤ Facebook + Email`, and `Social ≥ 2 × (SEO + AdWords)`.

The MIP solution allocated the full $10 M budget across four platforms — Print ($3 M), Facebook ($3 M), AdWords ($2.33 M), and LinkedIn ($1.67 M) — yielding a total expected return of $0.45 M and an overall ROI of 4.53 % (Figure 3). Compared with the LP result from Part 3 (ROI = 5.44 %), this outcome is slightly lower but far more practical: the integer formulation prevents fractional tier allocations and enforces valid tier sequencing. Overall, the model balances investment across print and digital media, producing a more defensible and implementable plan when ROI data are irregular.

Comparison of Part 3 (LP) and Part 4 (MIP) Solutions

To illustrate how the optimization strategy changed between concave and non-concave ROI structures, Figures 4 and 5 compare the LP (Part 3) and MIP (Part 4) outcomes.

In Figure 4, the grouped bar chart shows that the LP model concentrated spending on Email, Instagram, and TV, where ROI values decreased smoothly across tiers. Once integer logic was

introduced in the MIP model, the allocation diversified — shifting toward Print and Facebook ($3 M each), increasing AdWords to $2.33 M, and adding LinkedIn at $1.67 M. These shifts reflect the binary-activation logic, which prevents “jumping” to higher-ROI tiers without fully funding earlier ones.

Figure 5 plots ROI versus Investment for each platform under both models. The LP model achieved higher nominal ROIs (up to 6 %) because it was allowed to fractionally split across tiers. The MIP model produced slightly lower ROIs (≈ 4.5 %) but within a feasible tier sequence, creating a more realistic plan that respects managerial constraints and tier dependencies. Overall, the comparison highlights the trade-off between theoretical efficiency and operational feasibility — a critical insight for data-driven budget planning.

Figure 3:

```
==================== Part 4: MIP Solution (roi_company2.csv) ====================
Total Budget Used: $10.00M
Total Expected Return: $0.45M
Overall ROI: 4.53%

Platform Breakdown:
------------------------------------------------------------
AdWords      | Invested: $2.33M | Return: $0.11M | ROI:  4.56%
Facebook     | Invested: $3.00M | Return: $0.13M | ROI:  4.33%
LinkedIn     | Invested: $1.67M | Return: $0.07M | ROI:  4.43%
Print        | Invested: $3.00M | Return: $0.14M | ROI:  4.75%
============================================================
```

Figure 4:

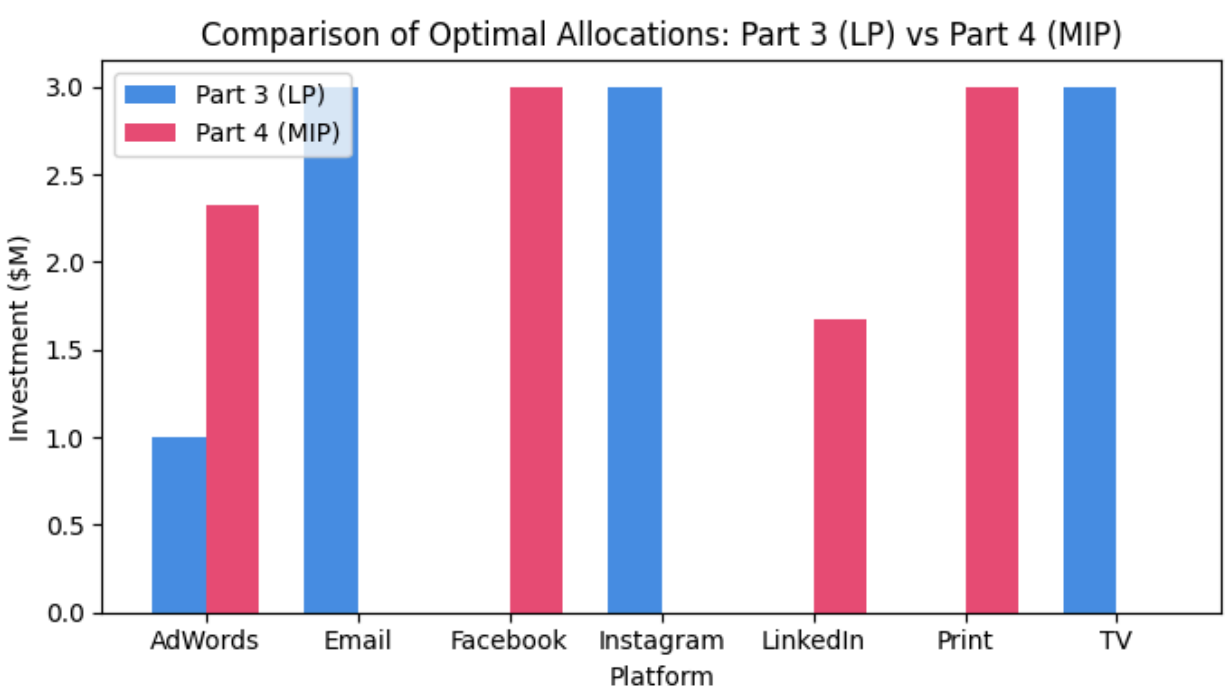
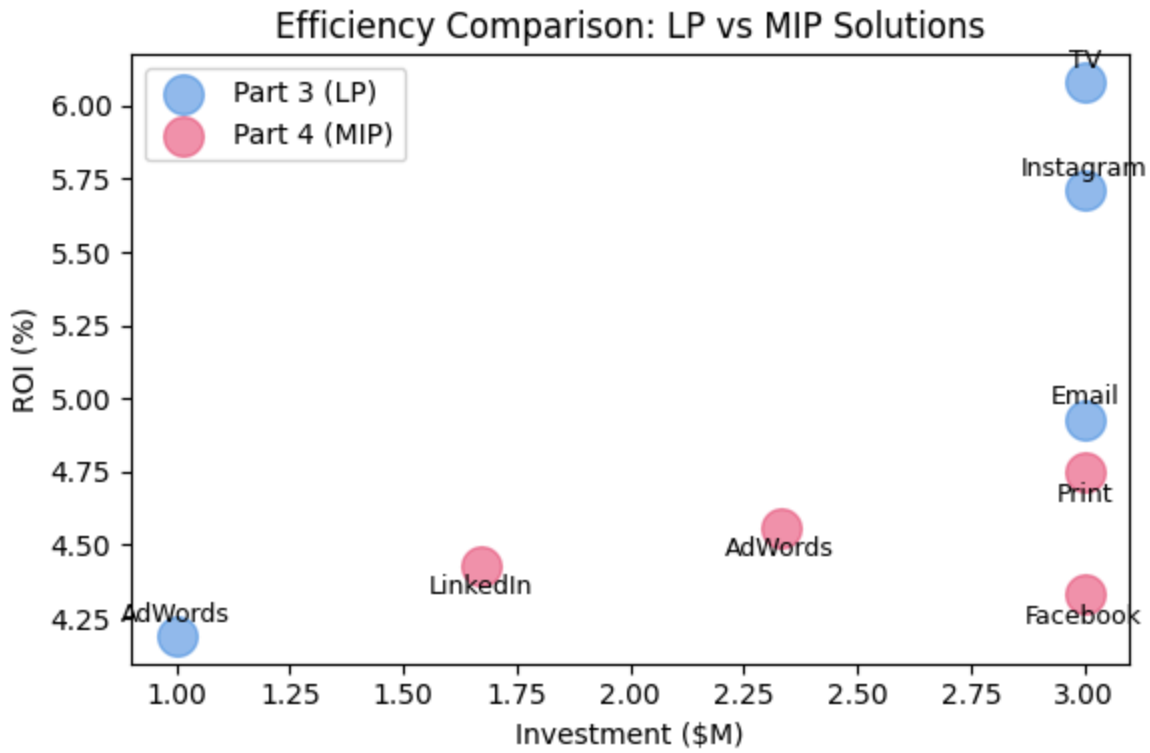


Figure 5:



Part 5:

In this stage, we assessed how sensitive the recommended media plan is to uncertainty in the ROI estimates by cross-evaluating each allocation on the other firm's ROI table. The LP plan from Part 3 (fit to Company-1 ROI) achieves 0.543640 on ROI1, while the MIP plan from Part 4 (fit to Company-2 ROI) achieves 0.452827 on ROI2. When we apply the Part-3 allocation to ROI2, the value drops to 0.277720, a loss of 0.175107 (−38.67% vs. the ROI2 optimum). Conversely, applying the Part-4 allocation to ROI1 yields 0.274903, a loss of 0.268737 (−49.43% vs. the ROI1 optimum). The allocations are not identical, reflecting different high-ROI tiers implied by each firm's estimates.

Practically, these results show that both plans are model-dependent, but the Part-3 (ROI1-fit) plan is more robust on a minimax-regret basis (worst-case loss ≈ 39% vs. ≈ 49%). If a single plan must be chosen before ROI uncertainty is resolved, the ROI1-fit allocation is the safer choice. The $3M per-platform cap (Constraint 3) remains useful: it limits concentration in any one channel and helps stabilize performance when ROI curves differ across sources. If additional robustness is desired, the team could incorporate a minimax-regret objective or a mild diversification penalty to further reduce downside under alternate ROI scenarios.

Part 6:

In this stage, preliminary research introduced minimum effective spends per medium from min_amount.csv. For each channel we must either not invest or invest at least its threshold. We embed this with binary use variables at the platform level and keep the tier-activation structure to respect the piecewise ROI so that earlier tiers are filled before later ones. The model

maximizes expected return under the same managerial constraints as before: a total 10 M budget, a 3 M per-platform cap, Print + TV ≤ Facebook + Email, and Social ≥ 2 × (SEO + AdWords). The objective correctly values piecewise returns by accumulating lower-tier returns and adding only the within-tier incremental return.

The optimal solution uses the full 10 M and concentrates spend in four media: Print (3.00 M), Facebook (3.00 M), AdWords (2.33 M), and LinkedIn (1.67 M), for a total expected return of 0.453 M and an overall ROI of 4.53 percent. All minimums are satisfied for active channels, while the others (TV, Email, Instagram, SEO, Snapchat, Twitter) optimally receive 0 under the ROI2 surface given their thresholds. Two policy constraints are binding at the optimum: Print + TV equals Facebook + Email (3.00 M = 3.00 M) and Social equals two times Search (4.667 M = 4.667 M), indicating they actively shape the allocation. Overall, the minimum-spend rule yields a practical on/off plan that remains consistent with the ROI2-driven priorities, concentrating budget where post-threshold returns are strongest and avoiding small, ineffective placements.

Part 7 – Monthly ROI with 50% Reinvestment (LP, month-by-month)

Setup : We extend the allocation problem over twelve months with rolling budgets. Each month begins with a base budget of $10M. On top of that, we add 50% of the *realized* return from the previous month. For example, if last month's return was $0.40M, next month's budget increases by $0.20M.

The file "roi_monthly.csv" contains, for every month and platform, piecewise ROI tiers (higher ROI for the first dollars, then step-downs). We apply the same managerial rules every month:

- Print + TV cannot exceed Facebook + Email.
- Total spend on Social (Facebook, LinkedIn, Instagram, Snapchat, Twitter) must be at least twice the spend on Search (SEO, AdWords).
- No single platform can receive more than $3M in a month.
- Spend in each tier cannot exceed that tier's width as given in the file.

Because ROIs are constant within a tier and only drop at known breakpoints, each month's problem is a linear program. We solve months sequentially, using that month's optimal return to update the following month's budget.

Model : For a given month, we choose how many dollars to place in each platform's ROI tiers. The objective is to maximize that month's total return (ROI × spend) subject to:

1. The full monthly budget is allocated.
2. Spend on any single platform is capped at $3M.

3. Print + TV is not allowed to exceed Facebook + Email.

4. Social spend is at least twice Search spend.

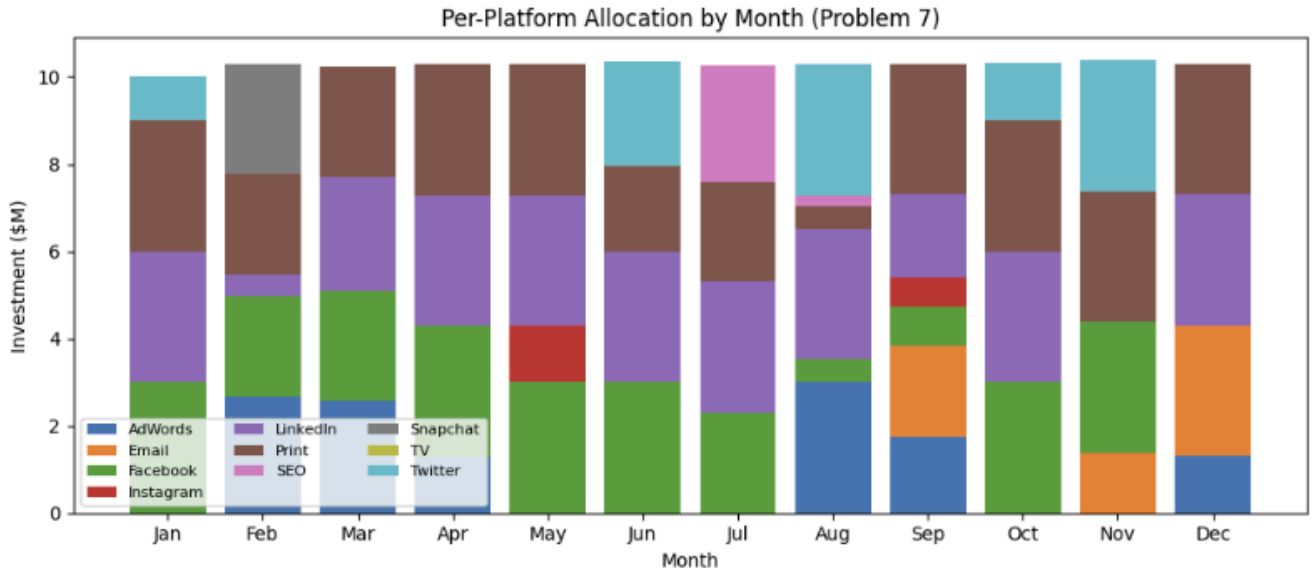
5. Each tier has a hard cap equal to its tier width.

After we compute the optimal return for the month, we set next month's budget to $10M plus half of that return.

Results : The optimizer concentrates dollars in the top ROI tiers while keeping the mix within the business rules. Budgets evolve moderately over the year—from $10.00M at the start to roughly the $10.3–$10.4M range at the peak—reflecting monthly returns in the $0.45–$0.76M band given the cross-channel constraints and platform caps.

As an illustration, in January the entire $10M is placed on a small set of high-ROI channels (e.g., Facebook $3M, LinkedIn $3M, Print $3M, Twitter $1M). This mix satisfies both the Print/TV vs. Facebook/Email balance and the Social-over-Search requirement. Month to month, the exact composition adjusts as the ROI tiers change, but no platform is ever allowed to exceed $3M.

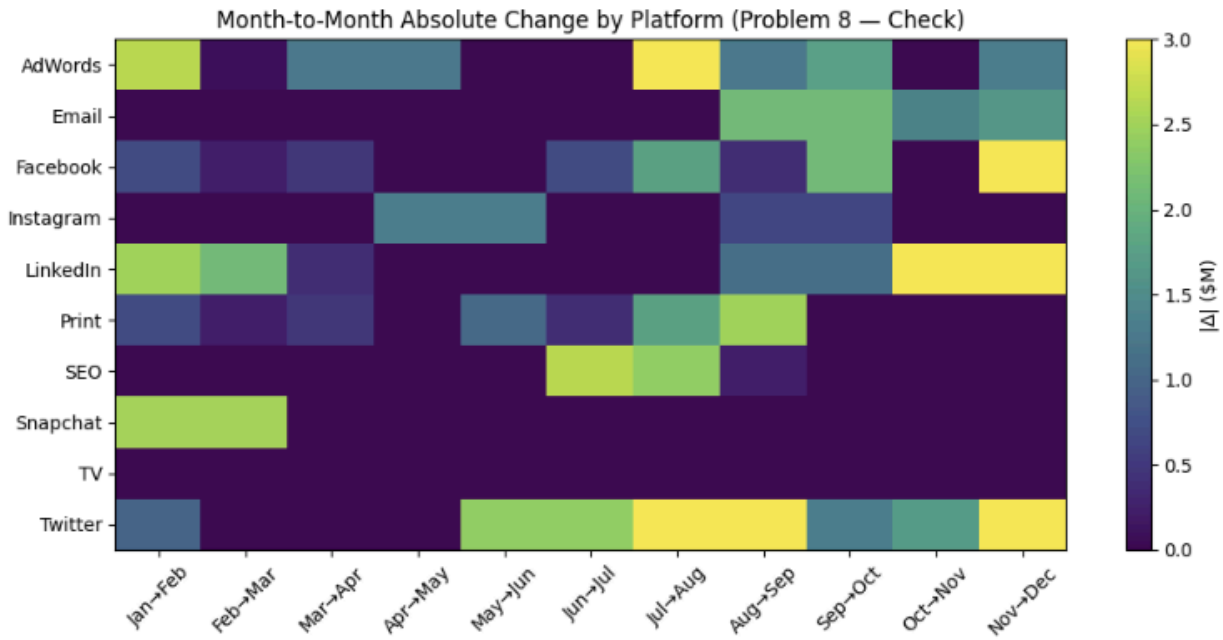
See Figure below for the month-by-month platform mix.



Managerial takeaway : Optimizing each month with a simple reinvestment rule yields agile plans that chase the best available tiers while respecting guardrails management cares about. The model is fully linear, fast to re-run when "roi_monthly.csv" updates, and easy to audit—making it practical for monthly planning and "what-if" analysis.

Part 8 - Stability Check and How to Model Stable Plans

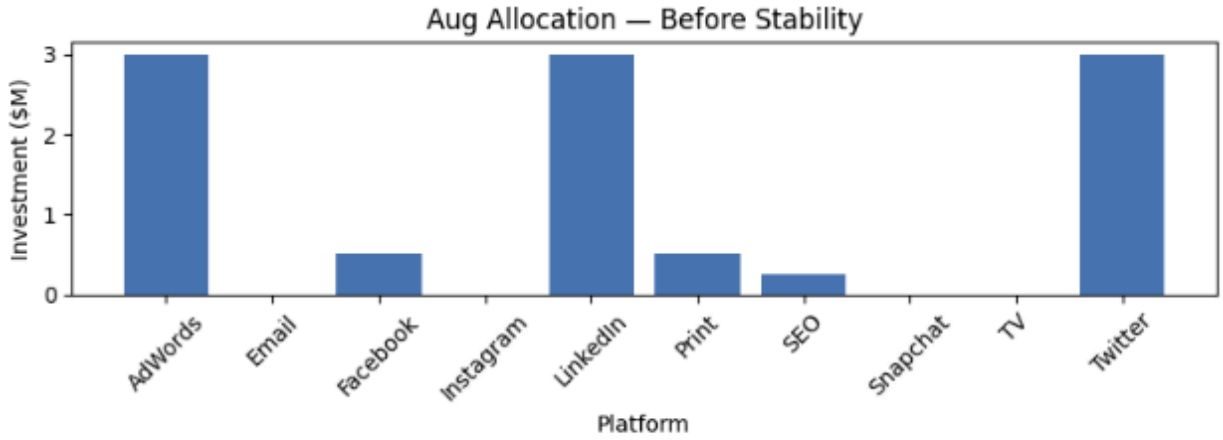
Diagnostic check : We evaluated the stability of the Part 7 plan using the project's definition: a plan is *stable* if for every platform, the month-to-month change in spend is ≤ $1M. Applying this rule to the month-by-month allocations from Part 7, we observed multiple violations (e.g., sharp increases on AdWords/Snapchat in early months and swings between Social and Search in late months). In other words, the purely return-maximizing rolling LP is not stable under the ±$1M rule. This is unsurprising: chasing the top tier each month often implies larger reallocations than a stability constraint would allow.

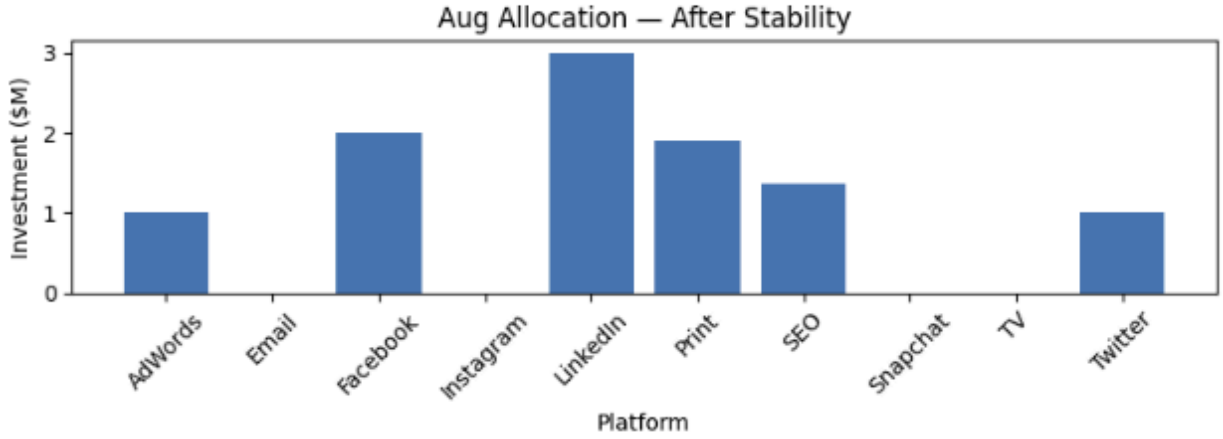


How to enforce stability (model design) : We enforce stability by limiting how much each platform's spend may change from one month to the next (e.g., within ±$1M). Practically, we carry forward last month's platform spends as fixed inputs, then introduce two nonnegative "change" variables per platform—one for how much it increases relative to last month, and one for how much it decreases. A single accounting constraint ties these to reality: this month's spend minus last month's spend equals "increase" minus "decrease." A second small bound caps their sum at the chosen threshold (e.g., $1M), which implements the absolute change limit without using absolute values and keeps the model linear.

All other rules remain exactly as before: the monthly budget must be fully allocated; no platform may exceed $3M; Print + TV cannot exceed Facebook + Email; Social must be at least twice Search; and each tier's width (from the ROI file) still limits spend. The result is a plan that trades a bit of return for smoother execution—fewer sudden swaps, steadier creative and trafficking, and more predictable delivery. The threshold is a tuning knob: tighter values (e.g., $0.5M) produce steadier but potentially lower-return plans; looser values (e.g., $1.5M) allow more agility.

After solving, we verify stability by computing month-to-month differences for every platform and confirming none exceed the threshold. A compact heatmap of absolute changes and a “violations per month” bar chart are effective to communicate where instability would have occurred without the constraint.





Implications : Enforcing stability introduces a controlled trade-off: total return may decrease slightly, but the plan becomes operationally smoother (easier flighting, reduced production/switchover costs, and more predictable audience delivery). If the team wants a tunable compromise, the ±$1M bound can be treated as a parameter (e.g., ±$0.5M for stricter smoothing; ±$1.5M for more flexibility), and we can report the revenue–stability frontier to stakeholders.

Conclusion

We started with a ten million dollar monthly budget and three non-negotiables from leadership: no platform above three million dollars, Print and TV cannot exceed Facebook and Email, and total Social must be at least twice Search. With the Company 1 ROI table, the linear model consistently filled the highest yielding early tiers while staying inside those rules. When we

tested Company 2, where the ROI tiers do not decline cleanly, a mixed-integer version enforced the correct tier order and delivered a feasible plan that respected the same business guardrails. Cross-testing the two ROI tables showed that a plan optimized for one table can underperform on the other, which argues for routine robustness checks whenever inputs change. Minimum effective spend rules also helped by turning platforms on only when the post-threshold return justified it.

Extending the exercise across twelve months with a simple reinvestment rule produced steady budget growth. Budgets moved from roughly ten million dollars at the start toward the low ten point four million range at the peak. Monthly returns stayed in a realistic band around one half million dollars. Month by month, the optimizer shifted dollars toward whichever tiers were best that month while keeping every business rule intact. The stacked allocation chart makes the mix changes obvious and easy to brief.

We then checked operational stability. Using the project's one million dollar month-to-month limit per platform, the unconstrained plan showed many jumps, especially when leadership flipped between Social and Search. The fix is straightforward and remains fully linear. Carry last month's platform spends, track this month's increases and decreases, and cap the total movement per platform at the chosen threshold. This converts the stability limit into a dial. Tightening it delivers smoother execution with a small loss of return. Loosening it allows more agility when the business needs it. The absolute-change heatmap and the violations-per-month bar chart make the benefit visible at a glance.

Recommendation

Use the linear program when tiers decline within each platform. Switch to the mixed-integer model when the tier order must be enforced. Adopt the rolling monthly framework so that half of realized return automatically feeds the next budget. Add an explicit stability cap on month-to-month platform changes. One million dollars per platform is a practical starting point that balances responsiveness and execution risk, and we can tune it tighter or looser based on flighting and production capacity. This approach delivers higher expected return within the guardrails leadership set, adapts as market conditions shift, and stays fast to re-run as ROI tables or constraints evolve.